

smaller, smarter and lower-priced medical devices and systems with more functionalities. Global demand of wearable, wireless and disposable medical sensors continues to rise.

In medical diagnostics, various sensors are required for digital blood pressure meters, digital thermometers, breath analysers, spirometers, peak-flow meters, respiration pulse oximeters and sleep apnea monitoring devices, among others.

Biomedical sensors

In medicine and biotechnology, biomedical sensors can detect specific biological, chemical or physical processes and then transmit or report data. These sensors can also be components in systems that process clinical samples, such as increasingly common lab-on-a-chip devices.

Biomedical sensors are also used to monitor the safety of medicines, food, environmental conditions and other substances we may encounter. Upgrades in patient treatments can come to fruition through distinguished proof of new markers of malady, quick findings and more successive or non-stop observing, in the healing centre.

The empowering innovation for these energising advancements ranges from sub-atomic tests to entire estimation frameworks, and envelops substance sensors and biosensors, low-power circuits for information preparing and remote transmission, and novel ongoing sign handling.

The ecosystem of medical sensors includes research and development, assembly and manufacturing, suppliers, system integrators, distribution, marketing and sales, and end users.

Medical sensors can be categorised on the basis of type, application and placement.

On the basis of type. Temperature sensors, blood glucose sensors, blood oxygen sensors, ECG sensors, image sensors, motion sensors, inertial sensors, pressure sensors

On the basis of application.



Fig. 2: DMI device



Fig. 3: Vital sign monitoring platform



Fig. 4: Eigen Lifescience device

Diagnostics, monitoring, medical therapeutics, imaging, wellness and fitness

On the basis of sensor placement. Strip sensors, wearable sensors, implantable sensors, invasive/non-invasive sensors, ingestible sensors

Special features of a biosensor are:

- Immobilised biological active material is used as a catalyst, and expensive reagents could be repeatedly used to detect same biological parameters.
- A biosensor has intensive specificity. Biomaterial only senses definitive ingredient and it is not affected by colour and concentra-

tion of measured material.

- A biosensor could quickly analyse the result of the measurand.
- A biosensor's accuracy is very high, and a relative error could reach one per cent.
- A biosensor's analysing system is very simple.
- Cost of a biosensor is low.

Next-generation biomedical sensors

DMI. DMI stands for DNA Medical Institute. DMI device is a portable, compact device that uses a single drop of blood to perform hundreds of clinical lab tests, telling patients with gold-standard accuracy whether they have anything from the common cold to Ebola. DMI has a wide reach, focusing on global health, emergency and critical care, and Space medicine.

Vital sign monitoring platform.

This device is strapped to the upper arm, where it measures a wide variety of data as patients go about their daily lives. Measurements include temperature, blood oxygenation, movement and activity, heart rate, and cutaneous blood perfusion and volume, among others. Data is then uploaded to the Cloud and viewed on a smartphone or computer.

Eigen Lifescience device. Eigen Lifescience device enables Hepatitis B rapid blood tests that can be analysed in minutes using the microprocessor on a smartphone. Designed for use in developing countries, the test can pinpoint patients who need treatment the most, thus, allocating services where these are most necessary. For instance, a child whose mother is infected with Hepatitis B must be treated within 12 hours. The experimental device might one day lead to testing for problems such as HIV, heart diseases and more.

Endotronix wireless health monitoring. Endotronix has patented a new type of wireless sensor reader. The experimental product is designed with a sensor that is